UNIT-4(File Management)
File Structure

LECTURE NO:-1

File

A file is a named collection of related information that is recorded on secondary storage such as magnetic disks, magnetic tapes and optical disks. In general, a file is a sequence of bits, bytes, lines or records whose meaning is defined by the files creator and user.

File Structure

A File Structure should be according to a required format that the operating system can understand.

- A file has a certain defined structure according to its type.
- A text file is a sequence of characters organized into lines.
- A source file is a sequence of procedures and functions.
- An object file is a sequence of bytes organized into blocks that are understandable by the machine.
- When operating system defines different file structures, it also contains the code to support these file structure. Unix, MS-DOS support minimum number of file structure.

File Type

File type refers to the ability of the operating system to distinguish different types of file such as text files source files and binary files etc. Many operating systems support many types of files. Operating system like MS-DOS and UNIX have the following types of files −

Ordinary files

- These are the files that contain user information.
- These may have text, databases or executable program.
- The user can apply various operations on such files like add, modify, delete or even remove the entire file.

Directory files

- These files contain list of file names and other information related to these files.

Special files

- These files are also known as device files.
- These files represent physical device like disks, terminals, printers, networks, tape drive etc.

These files are of two types −

- **Character special files** − data is handled character by character as in case of terminals or printers.
- **Block special files** − data is handled in blocks as in the case of disks and tapes.

File Access Mechanisms

File access mechanism refers to the manner in which the records of a file may be accessed. There are several ways to access files −

- Sequential access
- Direct/Random access
- Indexed sequential access

Sequential access

A sequential access is that in which the records are accessed in some sequence, i.e., the information in the file is processed in order, one record after the other. This access method is the most primitive one. Example: Compilers usually access files in this fashion.

Direct/Random access

- Random access file organization provides, accessing the records directly.
- Each record has its own address on the file with by the help of which it can be directly accessed for reading or writing.
- The records need not be in any sequence within the file and they need not be in adjacent locations on the storage medium.

Indexed sequential access

- This mechanism is built up on base of sequential access.
- An index is created for each file which contains pointers to various blocks.
- Index is searched sequentially and its pointer is used to access the file directly.

Space Allocation

Files are allocated disk spaces by operating system. Operating systems deploy following three main ways to allocate disk space to files.

- Contiguous Allocation
- Linked Allocation
- Indexed Allocation

Contiguous Allocation

- Each file occupies a contiguous address space on disk.
- Assigned disk address is in linear order.
- Easy to implement.
- External fragmentation is a major issue with this type of allocation technique.

Linked Allocation

- Each file carries a list of links to disk blocks.
- Directory contains link / pointer to first block of a file.
- No external fragmentation
- Effectively used in sequential access file.
- Inefficient in case of direct access file.

Indexed Allocation

- Provides solutions to problems of contiguous and linked allocation.
- A index block is created having all pointers to files.
- Each file has its own index block which stores the addresses of disk space occupied by the file.
- Directory contains the addresses of index blocks of files.

MTRC

UNIT-4(Directory structure)
Directory structure
LECTURE NO:-2

Directory structure

Several users create thousands and millions of files in a file system. To organize these files properly file system uses **directory structure**. The directory structure has the entries of the all the files related to that directory. We can consider that a directory is a **file,** that contains information about the other files.

Here, we will discuss some common schemes defining the directory structure and will also discuss some common maintenance operations that can be performed on directory structure. So let's start to explore directory structure.

Content: Directory Structure in OS

1. Overview of Directory Structure
2. Types of Directory Structures
3. Operations on Directory
4. Key Takeaways

Overview of Directory Structure

A File system contains thousands and millions of files, owned by several users. The directory structure organizes these files by keeping **entries** of all the related files. The file entries have information like file name, type, location, the mode in which the file can be accessed by other users in the system.

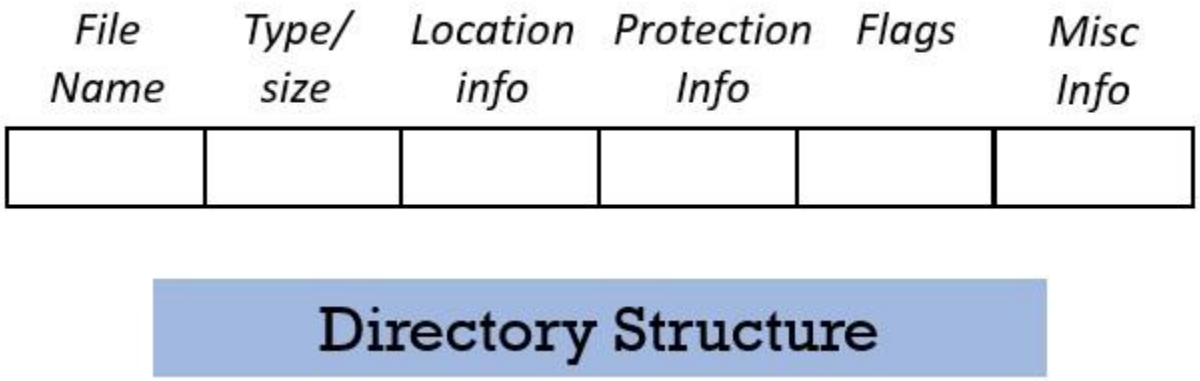
The two important features that must be taken care while maintaining a directory are given below.

1. A user must have freedom of giving the desired **name** to the file, without taking into consideration that the same file names are used by other users on the system.

2. A user must be able to **share** the files he creates and he must also be able to access the files, created by other users.

Directory structure provides both the above-discussed features. A directory always has information about the **group** of related files. Whenever a user or a process request for a file,

the file system search for the file's entry in the directory and when the match is found, it obtains the file's location from there.



Let us now study the directory structure with the help of the figure above. The field **File name**, contains the name of the concerned file in the directory, Type field indicates the kind or category of the file, Location Info field indicates the location where the file is stored.

Protection Info field contains the information whether the file can be accessed by the other user in the system or not. Flag field contains the kind of directory entry like value D in Flag field indicates that the file is a directory, value L indicates that the file is a link, value M indicates that the file is a mounted file system.

The Misc info filed in the directory contains the miscellaneous information about the owner of the file, the time of its creation, the time at which the file was modified last.

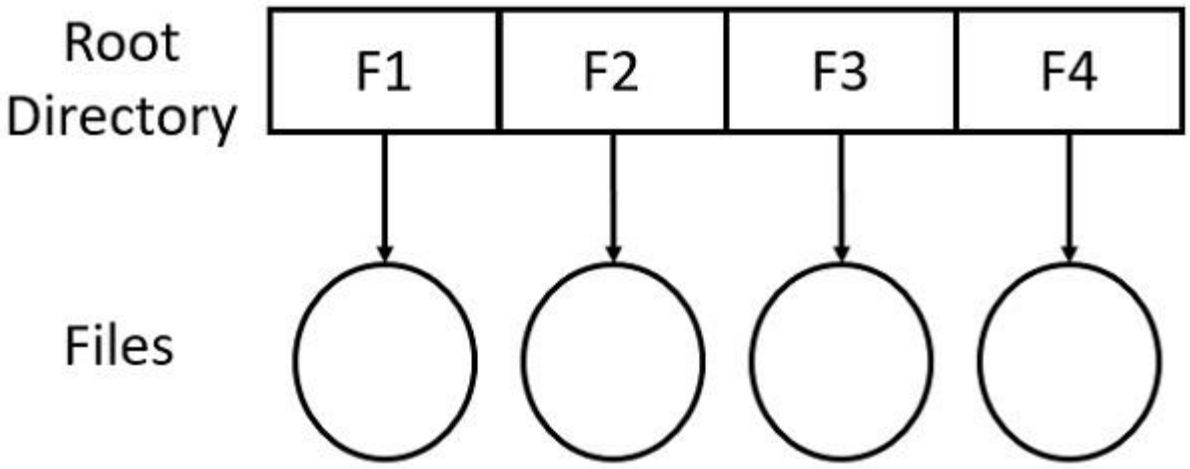
Types of Directory Structures

In this section, we will be discussing the most common types of directories, their benefits and drawbacks.

Single-level directory structure

Single level directory structure has only one directory which is called the **root** directory. The users are not allowed to create subdirectories under the root directory. All the files created by the several users are present in the root directory only.

As you can see in the diagram below all the file F1, F2, F3, F4 created by the different users are present at the root directory.

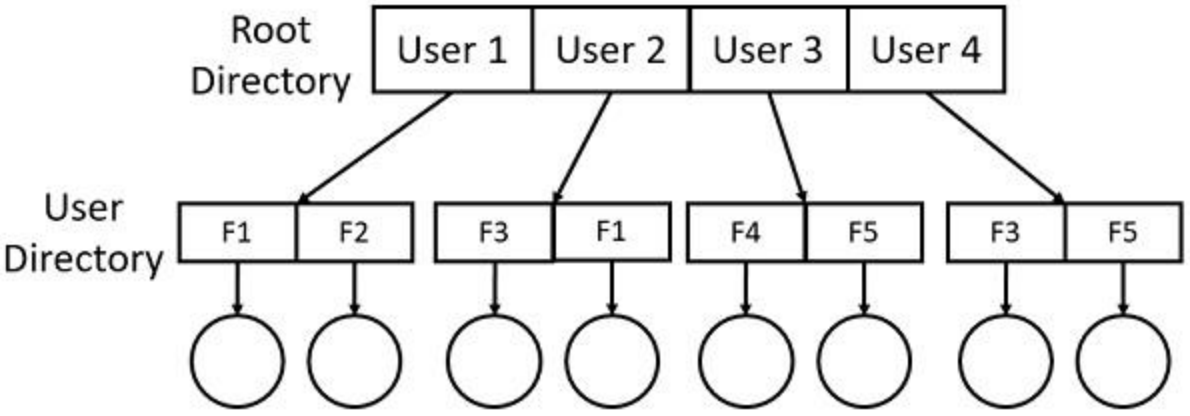


Single-level Directory Structure

There is one drawback of Single-level directory structure, a user **cannot use the same file name** used by another user in the system. Even if the file with the same name is created the old file will get destroyed first and replaced by the new file having the same name.

Two-level directory structure

In **Two-level directory structure,** the users create directory directly inside the root directory. But once a user creates such directory, further he cannot create any subdirectory inside that directory. Observe the figure below, 4 users have created their separate directory inside the root directory. But further, no subdirectory is created by the users.

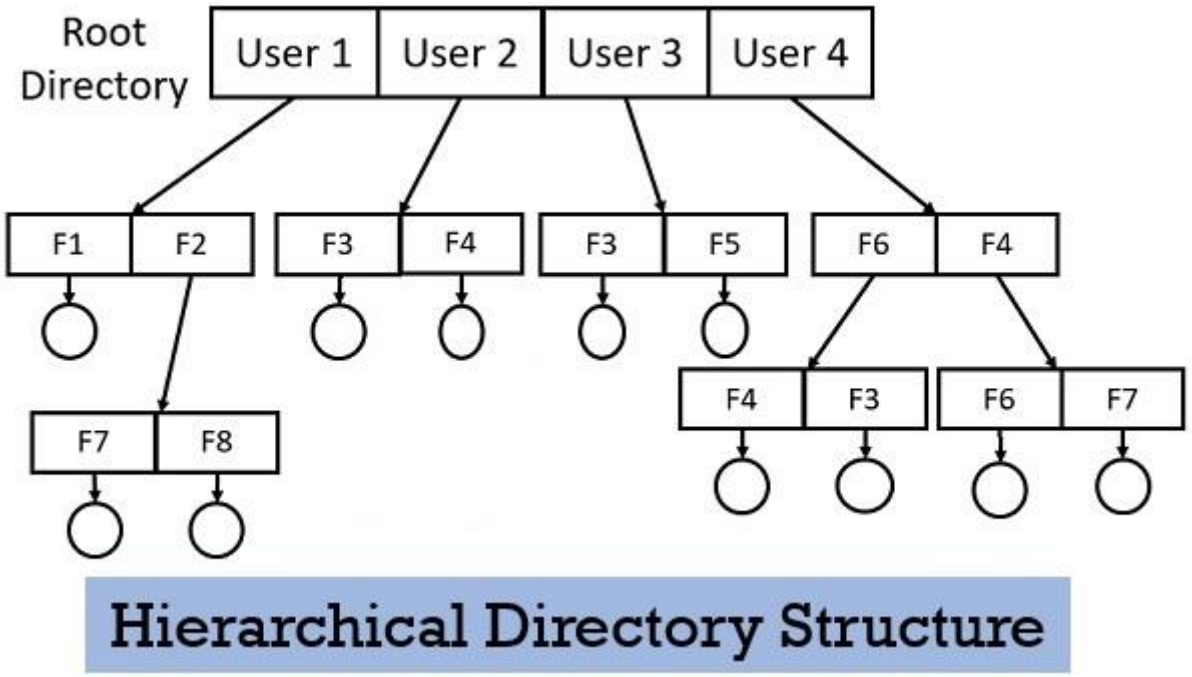


Two-level Directory Structure

This two-level structure allows each user to keep their files separately inside their own directory. This structure allows to use the same name for the files but under different user directories.

Hierarchical Directory Structure

In Hierarchical directory structure, the users can create directories under the root directory and can also create sub-directories under this structure. As the user is free to create many sub-directories, it can create different sub-directories for different file types.



Hierarchical Directory Structure

Here, the files are accessed by their location using the path. There are two types of paths to locate the file in this directory structure

Absolute Path

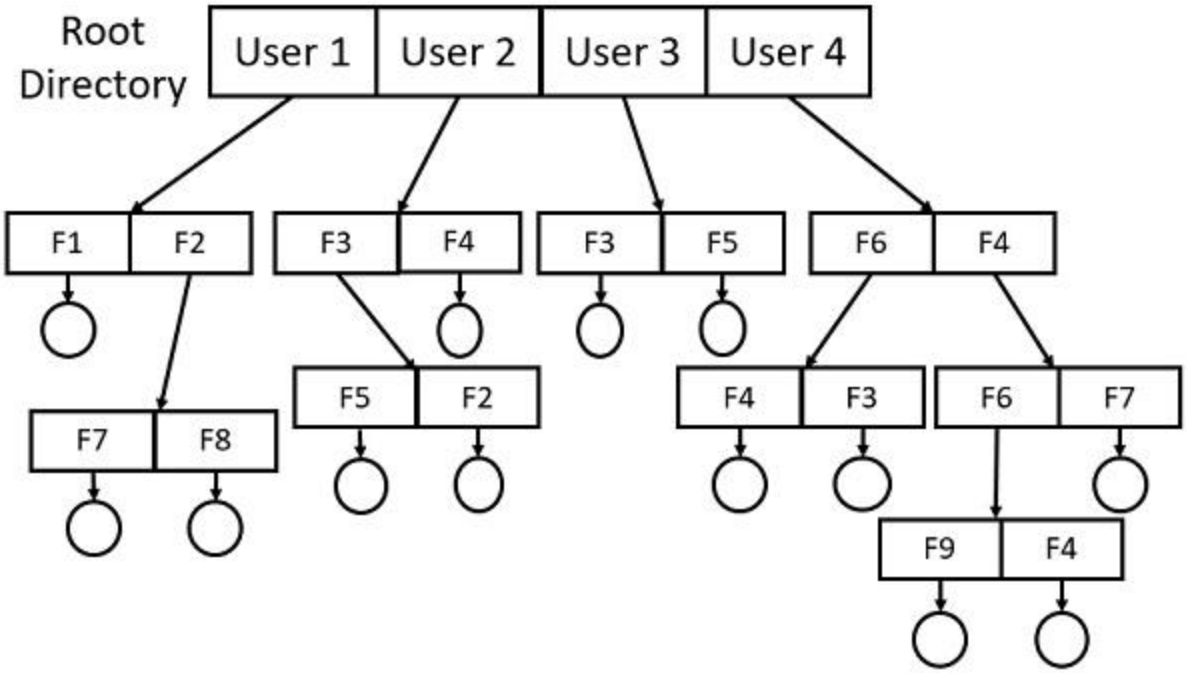
Here, the path for the desired file is described by considering the root directory as the base directory.

Relative Path

Here, either the user's directory is considered as the base directory or the desired file directory is considered as the base directory.

Tree Directory Structure

In a tree directory structure, except root directory, every directory or file has only **one parent directory**. So, there is a total separation between the users which provide complete naming freedom. Here, if a user wishes to access another users file, it has to go through two or more directories**.**



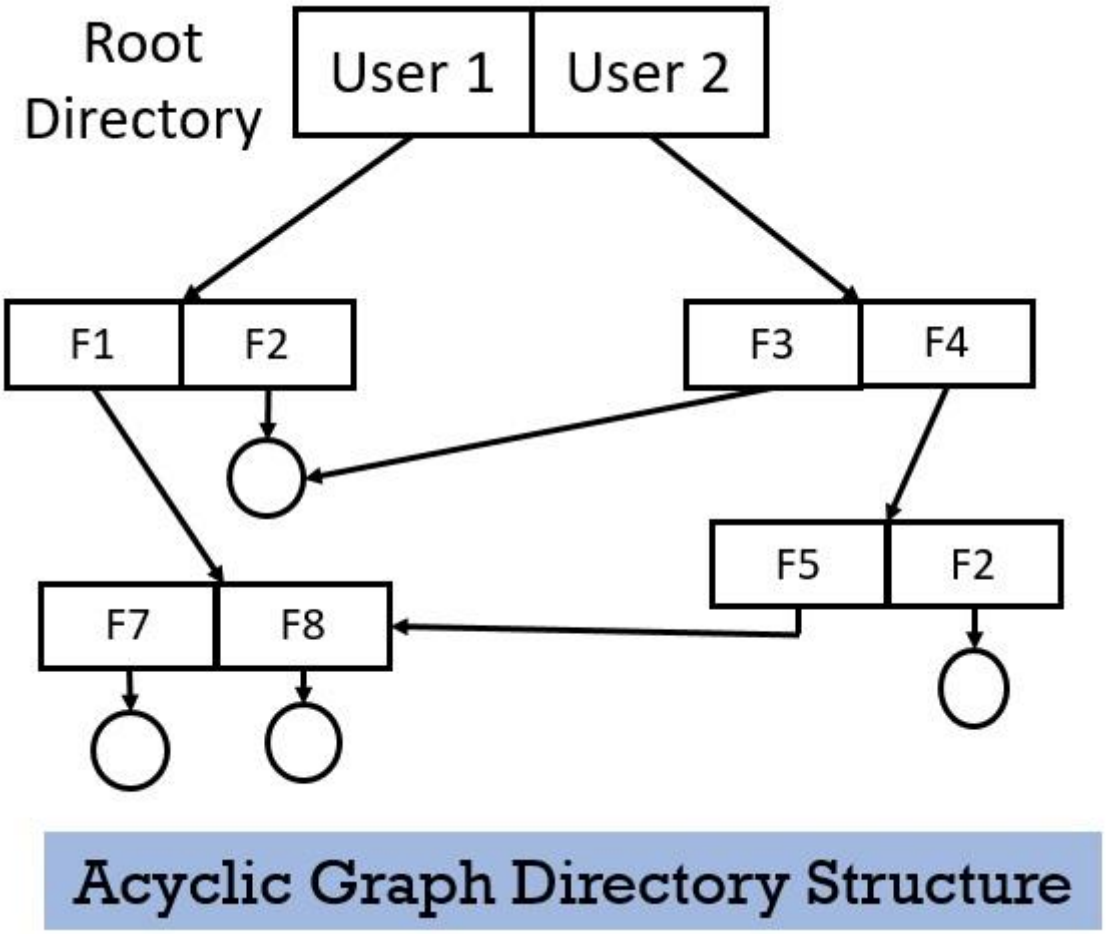
Tree Directory Structure

The tree directory structure provides an asymmetric way for the user to access the shared files of a different user. For example, a user can access a file of its own user directory with a shorter path than the other user.

Acyclic-Graph Directory Structure

This problem can be solved by the **acyclic-graph** directory structure. As this directory structure allows a directory or a file to have many parent directories. So, a shared file in a directory can be pointed by the other user directories who have access to that shared file using the links.

In the diagram below you can see that the directory having file F7 and F8 have two parent directories.



Acyclic Graph Directory Structure

Operations on Directory

As discussed above, a directory contains the entries of all the related files. For organizing the directory in the better way the user must be able to insert, delete, search, list the entries in the directory. Below we will be discussing the operation that can be performed on the directory.

1. Searching

A directory can be **searched** for a particular file or for another directory. It can also be searched to list all the files with the same name.

2. Creating

A new file can be created and inserted to the directory or new directory can be created keeping in mind that its name must be unique under that particular directory.

3. Deleting

If a file is no longer needed by the user, it can be deleted from the directory. The entire directory can also be deleted if it is not needed. An empty directory can also be deleted. When a directory is empty it is resembled by dot and dotdot.

4. List a directory

List of all the files in the directory can be retrieved and also the contents of the directory entry, for each file in a list. To read the list of all the files in the directory, it must be opened and after reading the directory must be closed to free up the internal tablespace.

5. Renaming

The name of the file or a directory represents the content it holds and its use. The file or directory can be renamed in case, the content inside or the use of file get changed. Renaming the file or directory also changes its position inside the directory.

6. Link

The file can be allowed to appear in more than one directory. Here, the system call creates a link between the file and the name specified by the path where the file is to appear.

7. Unlink

If the file is unlinked and is only present in one directory its directory entry is removed. If the file appears in multiple directories, only the link is removed.

Key Takeaways

- Directory is also considered as a file that contains information about the other files and directories.
- Directory keeps the entry of a group of all the related files and also the information like file names, types, location, etc.
- **The single-level** directory has only **one** root directory and no user is allowed to create subdirectories inside the root directory.
- **The two-level** directory allows the user to create **a directory** inside the root directory but do not allow creating further subdirectories.

- **Hierarchical** directory goes beyond two-level directory structure. Here, a user is allowed to create **many** subdirectories.
- In the **tree** directory, each directory has only **one parent** directory except the root directory.
- An **acyclic graph** structure, a directory can have **more** than one parent directory.
- The directories can undergo many maintenance operations like create, delete, list, rename, link, unlink, etc.

The directory structure is simply a file that keeps complete information of other files or directories in the file system.

MITRC

UNIT-4(File Access Methods)
File Access Methods

LECTURE NO:-3

File Access Methods in Operating System:

When a file is used, information is read and accessed into computer memory and there are several ways to access this information of the file. Some systems provide only one access method for files. Other systems, such as those of IBM, support many access methods, and choosing the right one for a particular application is a major design problem.

There are three ways to access a file into a computer system: Sequential-Access, Direct Access, Index sequential Method.

1. **SequentialAccess–**

   It is the simplest access method. Information in the file is processed in order, one record after the other. This mode of access is by far the most common; for example, editor and compiler usually access the file in this fashion.
   Read and write make up the bulk of the operation on a file. A read operation -*read next*- read the next position of the file and automatically advance a file pointer, which keeps track I/O location. Similarly, for the write*write next* append to the end of the file and advance to the newly written material.

   Key points:

   - Data is accessed one record right after another record in an order.
   - When we use read command, it move ahead pointer by one
   - When we use write command, it will allocate memory and move the pointer to the end of the file
   - Such a method is reasonable for tape.

2. **DirectAccess–**

   Another method is *direct access method* also known as *relative access method*. A filed-length logical record that allows the program to read and write record rapidly.

in no particular order. The direct access is based on the disk model of a file since disk allows random access to any file block. For direct access, the file is viewed as a numbered sequence of block or record. Thus, we may read block 14 then block 59 and then we can write block 17. There is no restriction on the order of reading and writing for a direct access file.

A block number provided by the user to the operating system is normally a *relative block number*, the first relative block of the file is 0 and then 1 and so on.

3. **Index sequential method–**

It is the other method of accessing a file which is built on the top of the direct access method. These methods construct an index for the file. The index, like an index in the back of a book, contains the pointer to the various blocks. To find a record in the file, we first search the index and then by the help of pointer we access the file directly.

Key points:

- It is built on top of Sequential access.
- It control the pointer by using index.

Access matrix in Operating System

Access Matrix is a security model of protection state in computer system. It is represented as a matrix. Access matrix is used to define the rights of each process executing in the domain with respect to each object. The rows of matrix represent domains and columns represent objects. Each cell of matrix represents set of access rights which are given to the processes of domain means each entry(i, j) defines the set of operations that a process executing in domain Di can invoke on object Oj.

	F1	F2	F3	PRINTER
D1	Read		read	

	F1	F2	F3	PRINTER
D2				print
D3		read	execute	
D4	read write		read write	

According to the above matrix: there are four domains and four objects- three files(F1, F2, F3) and one printer. A process executing in D1 can read files F1 and F3. A process executing in domain D4 has same rights as D1 but it can also write on files. Printer can be accessed by only one process executing in domain D2. The mechanism of access matrix consists of many policies and semantic properties. Specifically, We must ensure that a process executing in domain Di can access only those objects that are specified in row i.

Policies of access matrix concerning protection involve which rights should be included in the (i, j)th entry. We must also decide the domain in which each process executes. This policy is usually decided by the operating system. The Users decide the contents of the access-matrix entries.

Association between the domain and processes can be either static or dynamic. Access matrix provides an mechanism for defining the control for this association between domain and processes. When we switch a process from one domain to another, we execute a switch operation on an object(the domain). We can control domain switching by including domains among the objects of the access matrix. Processes should be able to switch from one domain (Di) to another domain (Dj) if and only is a switch right is given to access(i, j).

	F1	F2	F3	PRINTER	D1	D2	D3	D4
D1	read		Read			switch		
D2				print			switch	switch
D3		read	execute					
D4	read write		read write		switch			

According to the matrix: a process executing in domain D2 can switch to domain D3 and D4. A process executing in domain D4 can switch to domain D1 and process executing in domain D1 can switch to domain D2.

NITRC

UNIT-4(File Security)
File Security
LECTURE NO:-4

File Security

Security refers to providing a protection system to computer system resources such as CPU, memory, disk, software programs and most importantly data/information stored in the computer system. If a computer program is run by an unauthorized user, then he/she may cause severe damage to computer or data stored in it. So a computer system must be protected against unauthorized access, malicious access to system memory, viruses, worms etc. We're going to discuss following topics in this chapter.

- Authentication
- One Time passwords
- Program Threats
- System Threats
- Computer Security Classifications

Authentication

Authentication refers to identifying each user of the system and associating the executing programs with those users. It is the responsibility of the Operating System to create a protection system which ensures that a user who is running a particular program is authentic. Operating Systems generally identifies/authenticates users using following three ways −

- **Username / Password** − User need to enter a registered username and password with Operating system to login into the system.
- **User card/key** − User need to punch card in card slot, or enter key generated by key generator in option provided by operating system to login into the system.
- **User attribute - fingerprint/ eye retina pattern/ signature** − User need to pass his/her attribute via designated input device used by operating system to login into the system.

One Time passwords

One-time passwords provide additional security along with normal authentication. In One-Time Password system, a unique password is required every time user tries to login into the system. Once a one-time password is used, then it cannot be used again. One-time password are implemented in various ways.

- **Random numbers** − Users are provided cards having numbers printed along with corresponding alphabets. System asks for numbers corresponding to few alphabets randomly chosen.
- **Secret key** − User are provided a hardware device which can create a secret id mapped with user id. System asks for such secret id which is to be generated every time prior to login.
- **Network password** − Some commercial applications send one-time passwords to user on registered mobile/ email which is required to be entered prior to login.

Program Threats

Operating system's processes and kernel do the designated task as instructed. If a user program made these process do malicious tasks, then it is known as **Program Threats**. One of the common example of program threat is a program installed in a computer which can store and send user credentials via network to some hacker. Following is the list of some well-known program threats.

- **Trojan Horse** − Such program traps user login credentials and stores them to send to malicious user who can later on login to computer and can access system resources.
- **Trap Door** − If a program which is designed to work as required, have a security hole in its code and perform illegal action without knowledge of user then it is called to have a trap door.
- **Logic Bomb** − Logic bomb is a situation when a program misbehaves only when certain conditions met otherwise it works as a genuine program. It is harder to detect.
- **Virus** − Virus as name suggest can replicate themselves on computer system. They are highly dangerous and can modify/delete user files, crash systems. A virus is generatlly a small code embedded in a program. As user accesses the program, the virus starts getting embedded in other files/ programs and can make system unusable for user

System Threats

System threats refers to misuse of system services and network connections to put user in trouble. System threats can be used to launch program threats on a complete network called as program attack. System threats creates such an environment that operating system resources/ user files are misused. Following is the list of some well-known system threats.

- **Worm** − Worm is a process which can choked down a system performance by using system resources to extreme levels. A Worm process generates its multiple copies where each copy uses system resources, prevents all other processes to get required resources. Worms processes can even shut down an entire network.
- **Port Scanning** − Port scanning is a mechanism or means by which a hacker can detects system vulnerabilities to make an attack on the system.
- **Denial of Service** − Denial of service attacks normally prevents user to make legitimate use of the system. For example, a user may not be able to use internet if denial of service attacks browser's content settings.

S.N.	Classification Type & Description
1	**Type A** Highest Level. Uses formal design specifications and verification techniques. Grants a high degree of assurance of process security.
2	**Type B** Provides mandatory protection system. Have all the properties of a class C2 system. Attaches a sensitivity label to each object. It is of three types. • **B1** − Maintains the security label of each object in the system. Label is used for making decisions to access control. • **B2** − Extends the sensitivity labels to each system resource, such as storage objects, supports covert channels and auditing of events. • **B3** − Allows creating lists or user groups for access-control to grant access or revoke access to a given named object.

3	**Type C** Provides protection and user accountability using audit capabilities. It is of two types. • **C1** − Incorporates controls so that users can protect their private information and keep other users from accidentally reading / deleting their data. UNIX versions are mostly Cl class. • **C2** − Adds an individual-level access control to the capabilities of a Cl level system.
4	**Type D** Lowest level. Minimum protection. MS-DOS, Window 3.1 fall in this category.

Computer Security Classifications

As per the U.S. Department of Defense Trusted Computer System's Evaluation Criteria there are four security classifications in computer systems: A, B, C, and D. This is widely used specifications to determine and model the security of systems and of security solutions. Following is the brief description of each classification.

Protection in File System

In computer systems, alot of user's information is stored, the objective of the operating system is to keep safe the data of the user from the improper access to the system. Protection can be provided in number of ways. For a single laptop system, we might provide protection by locking the computer in a desk drawer or file cabinet. For multi-user systems, different mechanisms are used for the protection.

Types of Access:

The files which have direct access of the any user have the need of protection. The files which are not accessible to other users doesn't require any kind of protection. The mechanism of the protection provide the facility of the controlled access by just limiting the types of access to the file. Access can be given or not given to any user depends on

several factors, one of which is the type of access required. Several different types of operations can be controlled:

- **Read –**

  Reading from a file.
- **Write –**

  Writing or rewriting the file.
- **Execute –**

  Loading the file and after loading the execution process starts.
- **Append –**

  Writing the new information to the already existing file, editing must be end at the end of the existing file.
- **Delete –**

  Deleting the file which is of no use and using its space for the another data.

- **List –**

  List the name and attributes of the file.

Operations like renaming, editing the existing file, copying; these can also be controlled. There are many protection mechanism. each of them mechanism have different advantages and disadvantages and must be appropriate for the intended application.

Access Control :

There are different methods used by different users to access any file. The general way of protection is to associate *identity-dependent access* with all the files and directories an list called access-control list (ACL) which specify the names of the users and the types of access associate with each of the user. The main problem with the access list is their length. If we want to allow everyone to read a file, we must list all the users with the read access. This technique has two undesirable consequences:
Constructing such a list may be tedious and unrewarding task, especially if we do not know in advance the list of the users in the system.

Previously, the entry of the any directory is of the fixed size but now it changes to the variable size which results in the complicates space management. These problems can be resolved by use of a condensed version of the access list. To condense the length of the access-control list, many systems recognize three classification of users in connection with each file

- **Owner –**

  Owner is the user who has created the file.
- **Group –**

  A group is a set of members who has similar needs and they are sharing the same file.
- **Universe –**

  In the system, all other users are under the category called universe.

The most common recent approach is to combine access-control lists with the normal general owner, group, and universe access control scheme. For example: Solaris uses the three categories of access by default but allows access-control lists to be added to specific files and directories when more fine-grained access control is desired.

Other Protection Approaches:

The access to any system is also controlled by the password. If the use of password are is random and it is changed often, this may be result in limit the effective access to a file. The use of passwords has a few disadvantages:

- The number of passwords are very large so it is difficult to remember the large passwords.
- If one password is used for all the files, then once it is discovered, all files are accessible; protection is on all-or-none basis.

UNIT-4(User Authentication)
User Authentication
LECTURE NO:-5

User Authentication in Operating System

Access Authorization

Authorization is the process of giving someone permission to do or have something. In multi-user computer systems, a system administrator defines for the system which users are allowed access to the system and what privileges of use (such as access to which file directories, hours of access, amount of allocated storage space, and so forth).

Assuming that someone has logged in to a computer operating system or application, the system or application may want to identify what resources the user can be given during this session.

Thus, authorization is sometimes seen as both the preliminary setting up of permissions by a system administrator and the actual checking of the permission values that have been set up when a user is getting access.

- Authorization is a process by which a server determines if the client has permission to use a resource or access a file.
- Authorization is usually coupled with authentication so that the server has some concept of who the client is that is requesting access.

- The type of authentication required for authorization may vary; passwords may be required in some cases but not in others.
- In some cases, there is no authorization; any user may be use a resource or access a file simply by asking for it. Most of the web pages on the Internet require no authentication or authorization.

Most web security systems are based on a two-step process. The first step is authentication, which ensures about the user identity and the second stage is authorization, which allows the user to access the various resources based on the user's identity. Modern operating systems depend on effectively designed authorization processes to facilitate application deployment and management.

Key factors contain user type, number and credentials, requiring verification and related actions and roles.

Access control in computer systems and networks relies on access policies and it is divided into two phases:

1. Policy definition phase where access is authorized.
2. Policy enforcement phase where access requests are permitted or not permitted.

Thus authorization is the function of the policy definition phase which precedes the policy enforcement phase where access requests are permitted or not permitted based on the previously defined authorizations.

Access control also uses authentication to check the identity of consumers. When a consumer attempts to access a resource, the access control process investigates that the consumer has been authorized to use that resource. Authorization services are implemented by the Security Server which can control access at the level of individual files or programs.

Authentication in Operating System

Authentication mechanism determines the user's identity before revealing the sensitive information. It is very crucial for the system or interfaces where the user's priority is to protect the confidential information. In the process, the user makes a provable claim about individual identity (his or her) or an entity's identity.

The credentials or claim could be a username, password, fingerprint etc. The authentication and non-repudiation, kind of issues are handled in the application layer. The inefficient authentication mechanism could significantly affect the availability of the service.

Use of Authentication in OS

- Authentication is used by a server when the server needs to know exactly who is accessing their information or site.
- Authentication is used by a client when the client needs to know that the server is system it claims to be.

- In authentication, the user or computer has to prove its identity to the server or client.
- Usually, authentication by a server entails the use of a user name and password. Other ways to authenticate can be through cards, retina scans, voice recognition, and fingerprints.
- Authentication by a client usually involves the server giving a certificate to the client in which a trusted third party such as Verisign or Thawte states that the server belongs to the entity (such as a bank) that the client expects it to.
- Authentication does not determine what tasks the individual can do or what files the individual can see. Authentication merely identifies and verifies who the person or system is.

Example of Authentication:

For example, there is a sender A sending an electronic document to the receiver B over the internet. How does the system will identify that the sender A has sent a message dedicated to the receiver B. An intruder C may intercept, modify and replay the document in order trick or steal the information this type of attack is called **fabrication**.

In the given situation authentication mechanism ensures two things; first, it ensures that the sender and receiver are righteous people and it's known as **data-origin authentication**. Secondly, it ensures the security of the established connection between sender and receiver with the help of secret session key so that it could not be inferred and it is known as **peer entity authentication**.

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